

al chemical composition that makes it electrically conductive. So it will show as moissanite on a two-in-one tester/multi-tester. Fortunately, type II diamond testers are now available but cost a few extra bucks.

Role of UV light

Some diamond testers (Fig. 3) have a built-in ultraviolet (UV) light source, and this has led to the logical assumption that UV light can be used for testing diamonds. Note that, there is absolutely no way we



Fig. 2: An electronic diamond tester



Fig. 3: Diamond tester with built-in UV light source

can distinguish a diamond from a non-diamond using UV light.

UV light does, however, have some use for professional gemologists because it can give an indication of probability when comparing a natural diamond with a synthetic one. And, UV light makes a difference when testing a moissanite using a moissanite tester or two-in-one tester/multi-tester. Moissanite testers that do not work on some odd moissanites work perfectly when the stone is exposed to UV light. This is because, according to the laws of Quantum Physics, UV light can make a noticeable difference to electrical conductivity.

Gem tester electronics

Fig. 4 shows the reference system diagram of a gem tester that uses high voltage to quickly test a gemstone for legitimacy. This basic concept is adaptable for determining a gem type based on its electrical conductivity.

In principle, an electronic circuit (including the gem under test) as part of a circuit path is used to measure its electrical conductivity and, hence, the gem type. The procedure of determining whether a gem is moissanite or synthetic diamond involves providing a high voltage across a gem surface, greater

than a breakdown voltage, and measuring a minuscule current that flows through the gem.

Here, the first (probe) and second (clip) contacts of the circuit couple the high voltage to the gem under test, while a low-impedance detector circuit is used to flag when the contacts are erroneously contacting each other during measurement. The amount of current flowing through the device is measured using a high-impedance resistor referenced to ground. Voltage on the resistor is measured and compared to a predetermined threshold voltage. Measurements above the threshold voltage indicate that the gem is conductive, and that the gem therefore is a synthetic diamond or moissanite.

Conversely, measurements below a threshold voltage indicate that practically no current flows through the gem, and that the gem therefore may be a diamond—if other tests such as a thermal conductivity test show it to be as one.

At the brain of the gem tester is a microcontroller for processing a multitude of analogue inputs to produce a number of outputs for ensuring whether a measurement for a gem under test is, in fact, within a specific range, signaling that it is moissanite or synthetic diamond.

Getting the feel

Using a diamond tester/moissanite tester/multi-tester is not complicated. A good start is to read its user guide/instruction manual line by line, and make sure you are not doing anything wrong. If your tester works consistently (gives the same reading on the same stone every time) and correctly (accurately identifies stones), then you do not need to know the principle on which it works. However, if you are having problems, then the following basic tips will help solve most problems:

1. The probe of the tester must be clean, especially if it has not

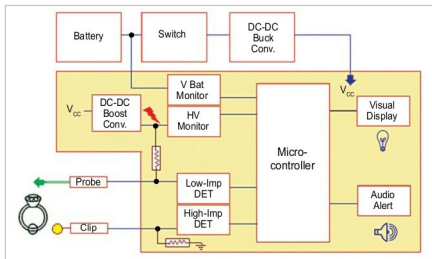


Fig. 4: System diagram of a gem tester